

Crop Recommendation System

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Abstract - Over 60% of India's workforce' is employed' in the husbandry sector, which contributes around 17% of the nation's GDP. This area has witnessed altercations bought about by modern technology, such as vertical farming, etc. Nonetheless, a large number of Indian farmers continue to use their land in accordance with customs and beliefs. For instance, instead of adapting to weather variations, they wait for the weather to suit their agricultural methods. By forecasting which crops are compatible with the variables that affect crop development, such as soil nutrient's, soil pH, humidity and rainfall our research aims to assist farmers in selecting the optimum crops for their circumstances and surroundings. We employ a variety of machine learning models, including Gaussian Naïve Bayes (GNB), Logistic Regression' (LR), Support Vector Machine' (SVM), and Decision Tree (DT), Random Forest Classifier.

Keywords - Crop suggestion, Nitrogen-Phosphorus Potassium (NPK), Humidity, Rainfall, pH, Machine Learning (ML), Decision Tree (DT), Support Vector Machine' (SVM), Logistic Regression (LR), and Gaussian Naïve Bayes (GNB), Random Forest Classifier.

1. INTRODUCTION

Agriculture' forms the backbone of the Indian finance, with Uttar Pradesh being a significant contributor to national food production. However, farmers often face challenges in making optimal decisions regarding crop selection, leading to potential losses in yield, inefficient resource utilization, and reduced profitability. Traditional farming practices, often based on experience and local knowledge, may not always account for the complex interplay of various factors such as soil characteristics, climate variability, market demands, and pest/disease prevalence.

The advent of data science and machine learning' offers promising solutions to address these challenges. By analyzing historical and real-time agricultural data, intelligent systems can be developed to provide personalized and data-driven recommendations to farmers regarding the most suitable crops to cultivate. This can lead to improved yields, optimized resource allocation (water, fertilizers), reduced risks associated with unfavorable conditions, and ultimately, enhanced economic returns for the farming community in Meerut and similar agricultural regions.

1.1 Background

Farmers have relied on various methods throughout history to predict which crops would be most successful:

- **Traditional Knowledge:** For centuries, farmers primarily depended on their accumulated experience, passed down through generations. This knowledge included:
 - i. *Observation of natural cycles:* Farmers observed the patterns of weather, seasons, and plant growth in their specific locations.
 - ii. *Soil assessment:* They learned to evaluate soil characteristics like texture, color, and drainage through simple methods.
 - iii. *Trial and error:* Through experimentation, farmers gradually identified which crops thrived in their fields and which did not.
- **Folklore and Cultural Practices:** In many cultures, folklore, and traditional practices played a role in agricultural

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decision-making. These practices might involve:

- i. *Astrological observations:* Some farmers followed lunar cycles or other celestial events to determine planting times.
 - ii. *Religious beliefs and rituals:* Certain crops or planting seasons might be associated with specific deities or ceremonies.
- **Early Scientific Methods:** As scientific understanding progressed, farmers began to incorporate more systematic approaches:
 - i. *Basic record-keeping:* Some farmers started keeping records of weather patterns, planting dates, and crop yields to identify trends.
 - ii. *Agricultural almanacs:* These publications provided information on weather forecasts, planting calendars, and farming tips.

1.2 Motivation

Several factors motivate the development of a crop recommendation system:

- **Increasing Population and Food Demand:** The growing global population necessitates increased agricultural productivity to ensure food security.
- **Soil Degradation:** Inappropriate crop selection and farming practices can lead to soil degradation, reducing its fertility and long-term productivity.
- **Need for Sustainable Agriculture:** There is a growing need for sustainable farming applications that reduce environmental pollution and save natural resources.

2. PROBLEM STATEMENT

Farmers in Uttar Pradesh, face the challenge of making informed decisions about crop selection due to the following factors:

Diverse soil types and varying fertility level across the region:-

- i. Unpredictable weather patterns and the risk of crop damage due to extreme weather events.
- ii. Lack of access to real-time information on optimal crop choices for their specific land and conditions.

These challenges often lead to:

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- a) Suboptimal crop yields and reduced farm income.
- b) Inefficient use of resources' such as water and fertilizers'.
- c) Increased risk of crop failure and financial losses for farmers.
- d) Negative environmental impacts due to unsustainable farming practices.

3. OBJECTIVES

The primary objectives of developing a crop recommendation system for Meerut, Uttar Pradesh, are to:

- Develop a system that can analyze relevant agricultural data, including soil data, climate data, and crop information, to provide accurate crop recommendations.
- Provide farmers with personalized crop recommendations tailored to their specific farm conditions, including soil type, weather patterns, and available resources.
- Develop a user-friendly interface that is easily accessible to farmers in Meerut, regardless of their technological expertise.
- Examine the performance of the system using appropriate metrics and validate its effectiveness in real-world agricultural settings.

4. LITERATURE REVIEW

There' are a several horticultural yield suggestion frameworks' accessible thinking regards different boundaries' by using ML calculations'. Different' ML method's are applied in agribusiness' area to concentrate on the authorised information' which can be needful to ranchers as well as country's finance.

Taj at el [1] issued both order and relapse procedures; to construct a harvest proposal framework. Information used is comprised of boundaries related soil condition' and climate conditions. Beginning characterization' is finished utilizing K-Nearest Neighbour (KNN) to find the best boundaries' that altogether make the harvest yield. Consequently, a relapse model using Artificial Neural Network's (ANN) is utilized to anticipate a harvest for proposal. This study was done particularly to address food security issue in Egypt. Banavlikar et al

Devadhe et al [2] presented a specialist' framework for crop determination considering boundaries like area wise month to month precipitation and efficiency of harvesting' during 2000 to 2014 in Maharashtra.

Linear regression, Decision tree and Random forest calculations are applied to anticipate the yield pace of harvests which might assist with working.

on the choice of grouping of occasional' harvests to be planted'. Be that as it may, because of thought of just two or three boundaries', the review is restricted exclusively for seasonl choice of harvests. Random Forest calculation' gave better forecast results when contrasted with straight relapse promotion choice tree.

Jain et al [3] proposed a harvest determination strategy in light of different elements like natural, financial and yield rate to boost the yield creation which can help ranchers and economy of country to defeat the food supply interest. To accomplish the expansion in yield, determination of harvest in view of specific elements assumes a significant part. The element of costing is added to different boundaries' like soil type, precipitation, temperature' for crop choice. Weka classifiers and relapse technique's are utilized to foresee the perfect determination of harvest and afterward a yield sequencing strategy is presented by utilizing crop sequencing calculation in view of yield rate and market cost. Additionally proposed AI models can be utilized in multiples ways in agribusiness area like water system, sickness identification, design discoveries which can additionally propel horticulture area. Rajak et al. [4] concocted a harvest proposal framework to boost crop yield by considering soil explicit properties gathered from soil testing research facilities from pune,

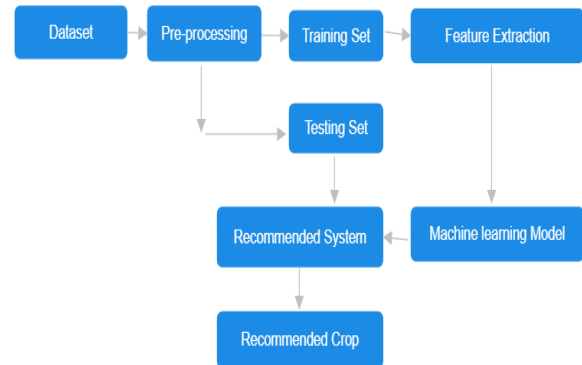
[5] Dighe, Deepti, Harshada Joshi, Aishwarya Katkar, Sneha Patil, and Shrikant' Kokate. "Survey of Crop Recommendation Systems". This proposed' system build a crop recommendation system for smart farming. In this research paper reviewed multiples machine learning algorithms' like CHAID, KNN, K- means, Decision Tree, Neural Network, Naïve Bayes, C4.5, LAD, IBK and SVM algorithms. For this research used Hadoop framework for the intensive calculations and also helped to get better accuracy for the system.

[6] Rajak, Rohit Kumar, Ankit Pawar, Mitalee Pendke, Pooja Shinde, Suresh Rathod, and Avinash Devare. "Crop recommendation system to maximize crop yield using machine learning technique". This proposed method is used for identifying particular crop based on soil database. This proposed system worked on various machine learning classifier like support vector machine (SVM) classifier, ANN classifier, Random Forest and Naïve Bayes for recommend a crop for site

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specific parameter with accuracy and efficiency.

5. PROPOSED WORK



In our model, we have presented a procedure that is separated' into various stages as appeared in Figure 1.

The five phases are as per the following:

- 1) Data Collection
- 2) Pre-processing' (Noise Removal)
- 3) Feature Extraction
- 4) Applied Machine Learning Algorithm
- 5) Recommendation System
- 6) Recommended Crop

5.1 Flow Of The Proposed System'

The system's workflow, visualized in the figure, details the progression from data to recommendation:

1. Data Collection: The dataset consists of parameter's like Nitrogen (N), Phosphorous (P), Potassium (K), PH value of soil, Humidity, Temperature and Rainfall. The datasets have been obtained from the Kaggle's site. For this study, a dataset featuring twenty-two different crop types was utilized, spanning cereals, pulses, fruits, and fibres. To ensure robust model evaluation, the dataset was divided into training and testing subsets. Specifically, 80% of the data was used to train the machine learning model, while 20% was used for independent performance assessment.

Dataset Link:

<https://www.kaggle.com/datasets/atharvaingle/crop-recommendation-dataset/data>

2. Data Pre-Processing: Data preprocessing is crucial for the success of our crop recommendation system. Data collected from various sources often arrives in a raw, inconsistent state, requiring careful refinement. This process

involves thorough data cleansing, where incomplete, redundant, and inconsistent data points are addressed through techniques like imputation and duplicate removal. Therefore, in this step such redundant data should be filtered. the common data preprocessing techniques used are Handling Missing Values, Outlier Detection and Treatment etc

3. Feature Extraction: The feature selection phase is crucial for optimizing the crop recommendation system's performance. It focuses on identifying and utilizing the most relevant attributes within the dataset, thereby eliminating extraneous and redundant information. This process aims to enhance model accuracy, reduce computational complexity, and improve interpretability. Irrelevant data, which exhibits minimal correlation with the target variable, is removed using statistical methods or domain expertise.

4. Testing Set and Recommended System: The trained Machine Learning Model is evaluated using the Testing Set, which contains data the model has not seen during training. This assesses the model's ability to generalize and make accurate predictions on new data.

The Recommended System utilizes the trained and evaluated Machine Learning Model. It takes new input data and applies the model to predict the most suitable crop.

5. Model Training: Training the selected machine learning model using the prepared historical data. This involves the algorithm learning the relationship's between the input features (soil, climate, etc.) and the output (suitable crop or yield).

6. Methodology: To examine the efficacy of various machine learning approaches for crop recommendation, this study explored the application of several algorithms. Specifically, Random Forest, Decision Tree, Support Vector Machine (SVM), Logistic Regression, Naive Bayes, k-Nearest Neighbours (k-NN), were implemented and their performance was compared to determine the most suitable model for the given dataset and problem.

7. Decision Tree: The Decision Tree algorithm was chosen for its ability to create easily interpretable models and to caught the non-linear relationships prevalent in agricultural data. To counteract the risk of overfitting, especially with complex trees, meticulous parameter tuning and pruning were applied, ensuring the model's capacity to generalize accurately to new data. This structured approach allowed for a clear understanding of the factors influencing crop recommendations, enhancing the system's overall reliability and usability.

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8. Support Vector Machine (SVM): Support Vector Machines (SVMs) is a supervised machine learning algorithm or model which can be utilized for classification and as well as for regression challenges.

By allowing input data into a higher-dimensional space, SVMs aim to get an optimal hyperplane that maximizes the margin between different crop classes. However, the computational cost associated with training SVMs on large datasets and the sensitivity to kernel selection required careful consideration.

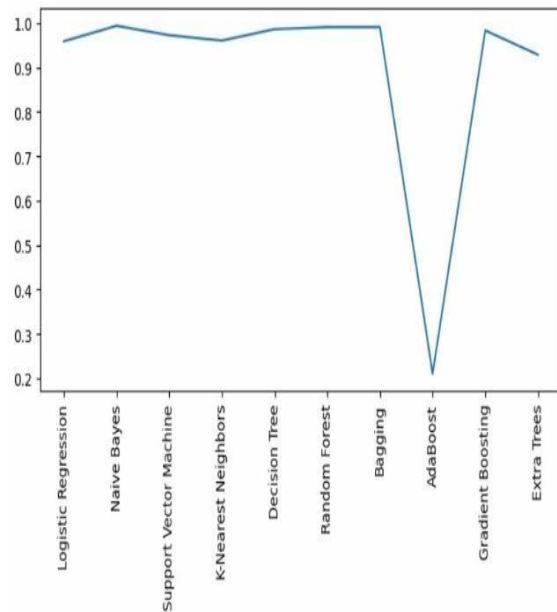
9. Logistic Regression: Logistic Regression, a widely used statistical method, was implemented to assess the linear separability of crop classes based on environmental and soil parameters. By modelling the chance of a crop belonging to a particular class using a logistic function, the algorithm provides a straightforward and interpretable approach to classification.

10. Random Forest Classifier: Random Forest Classifier is a type of ensemble algorithm. It produces a set of decision trees from a subset of a training set that is randomly picked. Greater the number of trees higher the accuracy in prediction. It overcomes overfitting problem by taking average of all the predictions cancelling out biases.

11. Gaussian Naive Bayes (GNB): Gaussian Naive Bayes (GNB) is a classification algorithm in machine learning that employs a probabilistic framework and the assumption of the gaussian distribution for continuous features. It operates on the principle of naïve independence, where each predictor variable is assumed to contribute independently to the prediction of target var .

6. RESULT AND FINDINGS

This study investigates the performance of multiple machine learning classifiers for a crop recommendation system. The accuracy of ten different algorithms – Logistic Regression, Naive Bayes, Support Vector Machine, K-Nearest Neighbours, Decision Tree, Random Forest, Bagging, AdaBoost, Gradient Boosting, and Extra Trees was evaluated. The results, visualized in the accompanying graph, indicate significant variations in the predictive capabilities of these models for crop recommendation. This paper presents a comparative analysis of these algorithms based on their observed accuracy, highlighting the strengths and weaknesses of each for this specific agricultural application.

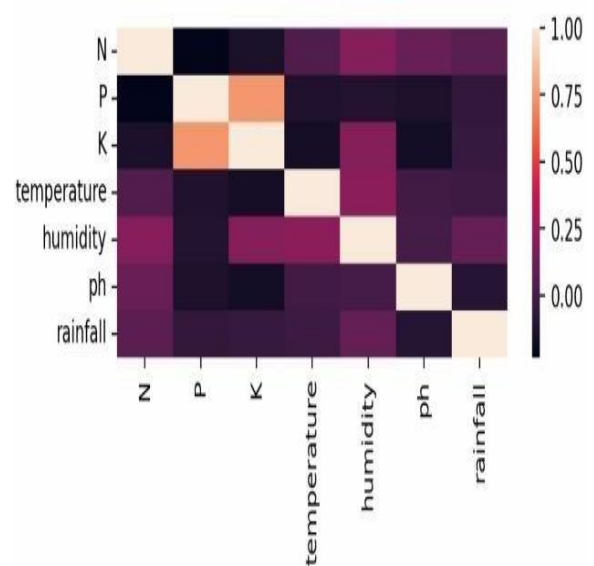


Therefore, based on this image, the reasons why we would take the **Random Forest classifier** for crop prediction are:

- High Accuracy/Performance:** The primary reason to choose any machine learning model is its ability to make accurate predictions. The Random Forest in this scenario demonstrates excellent performance, suggesting it can effectively classify the suitable crop based on the input features.
- Robustness to Overfitting:** Random Forests are known for their robustness to overfitting, especially compared to single decision trees. By averaging the predictions of multiple decision trees trained on different subsets of the data and features, they tend to generalize well to unseen data.
- Feature Importance:** Random Forests can provide information into the importance of different features in the prediction process. This can be valuable in understanding which factors (e.g., soil properties, climate data) are most influential in determining the recommended crop.
- Handles Non-linear Relationships:** Decision trees, the building blocks of Random Forests, can capture complex, non-linear relationships between the input features and the target variable (crop type). This is often the case in agricultural data.
- Less Sensitive to Outliers:** While not entirely immune, Random Forests tend to be less sensitive to outliers in the data compared to some other algorithms.

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This study investigates the linear relationships between key numerical features utilized in a crop recommendation system. Understanding the correlation patterns among these features – Nitrogen (N), Phosphorus (P), Potassium (K), temperature, humidity, pH, and rainfall – is crucial for feature selection, model interpretation, and potentially identifying multicollinearity issues that could affect the performance of machine learning models. A heatmap visualizing the Pearson correlation coefficients between these features is presented and analyzed. The findings reveal varying degrees of correlation, with some nutrient levels showing moderate positive correlation and environmental factors exhibiting relatively weak linear relationships.



7. CONCLUSION

This research developed and a Python-based Crop Recommendation System, leveraging the Random Forest Classifier, to deliver precise crop suggestions depend on critical soil and environmental factors. The system's capacity to process and interpret key agricultural variables such as Nitrogen, Phosphorus, Potassium, Temperature, Humidity levels, soil pH, and Rainfall proved highly effective in forecasting appropriate crop choices. Rigorous data preparation, including thorough cleaning and strategic feature engineering. The practical implications of this system are substantial, offering farmers opportunities to enhance crop output, minimize resource expenditure, promote environmentally sustainable agricultural practices, and make data-driven decisions. After evaluating several machine learning methods, we found that Random Forest Classifier had the highest accuracy. The resulting Python application serves as a user-friendly and readily accessible resource, empowering farmers to optimize their agricultural yields and operational efficiency.

8. FUTURE SCOPE

- Integrate real-time data from sensors' and weather' stations to provide dynamic crop recommendations.
- Develop a mobile application for easy access to the crop recommendation system.
- Expand the system to include economic factors (e.g., market prices, demand) to optimize crop selection for profitability.
- Explore the usage of others machine learning algorithms (e.g., deep learning models) to further improve prediction accuracy.
- Expand the model to include disease prediction.

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